

2024



Petal Scales Consumer Credit Access Using AWS

Learn how credit card startup Petal redefined consumer credit worthiness using AWS services.

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Improved scalability

to handle millions of credit card applications

Facilitated

quick iteration

Boosted

productivity

Maintained

high availability

Advanced mission

of democratizing



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Overview

Credit card startup Petal is a trailblazer in fintech history to qualify for Petal credit cards instantly. This transformative approach to credit underwriting, which Petal pioneered, has enabled hundreds of thousands of consumers who would typically be turned away by mainstream card issuers to gain access to credit in their lifetime.



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Petal was born in 2016 after one of its founders experienced challenges acquiring credit in the United States as an international student with no US credit history. This inspired the founders to reimagine the credit decisioning process to better serve those with little or no credit history.

Opportunity | Creating a Scalable, Reliable, and Mission-Aligned Credit Solution Using AWS



To fulfill the company's mission, Petal sought to build an infrastructure that could scale and grow with its business. Upon launching its first credit card, the Petal team needed to iterate quickly on its product. By leveraging AWS infrastructure, Petal's engineering team was able to focus on creating a smooth application journey and a compelling credit card experience for its customers. Multiple application journeys and data sources, including bank data, were integrated to maximize approval rates for applicants denied by more traditional issuers. Behavioral incentives, transparent payment options, spend information, and other features were added to the Petal app to encourage responsible credit use. By building on AWS scalable solutions, the engineering team was also able to avoid many of the scalability pitfalls faced by rapidly growing startups. When they did run into engineering challenges, they could simply leverage the AWS platform to scale out their infrastructure or reach out to AWS Support and their AWS account team.

"The uptime and scalability of our AWS stack facilitate our ability to focus on our company's mission," says John Wang, vice president of engineering at Petal. "It's been key to us being able to move quickly as a small startup and focus on delivering features and products that are important for our customers."

Solution | Using Amazon SageMaker to Train ML Models and Amazon Platform Services to Build and Scale Quickly

Petal's data-driven infrastructure runs on AWS services, from backend infrastructure to the front-office applications landscape. When customers apply for a credit card, they are brought to the web application to fill out required personal information. Petal hosts the user interface for these first webpages using [Amazon Simple Storage Service](#) (Amazon S3), an object storage service offering industry-leading scalability, data availability, security, and performance. "We want the first interaction to be very resilient," says Wang. "By using Amazon S3, we can maintain high availability of our application page for our millions of applicants."

For data storage, Petal uses [Amazon Relational Database Service](#) (Amazon RDS) [for PostgreSQL](#) to store its core transactional business data. Amazon RDS is a fully managed relational database that makes it simpler to set up, operate, and scale PostgreSQL databases in the cloud. RDS helped Petal quickly and easily scale its transactional data processing needs during periods of rapid growth. Petal also leverages [Amazon Redshift](#) and S3 for storing analytical and research data. Much of this data is fed into Petal's proprietary ML models used during underwriting and customer account management. To train its predictive models, Petal uses [Amazon SageMaker](#), which businesses can use to build, train, and deploy ML models for virtually any use case with fully managed infrastructure, tools, and workflows. SageMaker allowed Petal to iterate more quickly and effectively on its ML models, including the CashScore model, which is an integral part of Petal's underwriting and mission to better serve those with little to no credit history. "By using Amazon SageMaker, our data science team can independently govern and configure the kinds of instances needed to train their ML models," says Wang. "It has the capabilities they need."

Scalability has been crucial for Petal as its workflows and systems have grown more complex with the steady increase of users and applications. To handle the compute resource needs of its services and machine learning infrastructure, Petal uses [Amazon Elastic Compute Cloud](#) (Amazon EC2), which offers secure and resizable compute capacity for virtually any workload. To run its Kubernetes-based workloads on Amazon EC2 in an efficient manner, the company uses [Amazon Elastic Kubernetes Service](#) (Amazon EKS), a managed service to run Kubernetes in the cloud and on-premises data centers. On AWS, Petal's infrastructure easily scales vertically and horizontally as the company has grown from processing a few dozen applications in the beginning to now handling millions of applications. In addition to scalability, Amazon services help boost productivity for Petal engineers. "As our engineering team has gotten more sophisticated, we've needed better tooling," says Wang. "By turning to Amazon EKS, we take away a lot of the extra complexity of running and hosting Kubernetes natively for our small team."

Finally, Petal attributes its success building on AWS to the relationship its team has built with AWS. Through working proactively and collaboratively with its AWS team, Petal gains access to the latest technologies, critical expertise, unique programs, and valuable perspectives that have advanced its business-critical projects. Wang explains, "The AWS account and technical support staff have partnered closely with Petal to help us leverage all that AWS has to offer, including help troubleshooting technical issues, technical design reviews, and training in speaking and networking opportunities."



Outcome | Opening Up More Opportunities to Credit Access

Petal continues to expand its reach to consumers. After initially launching with one credit card, Petal now offers three cards, Petal 2, Petal 1, and Petal 1 Rise. This suite of products enables Petal to serve customers with a range of credit profiles and needs. Regardless of which card is in their wallet, Petal’s goal is the same for all customers: to empower those who have historically been overlooked by the traditional credit system and help them build a healthy financial future.

To date, using AWS, Petal has facilitated the approval of nearly 400,000 consumers for Petal cards. More than 40 percent of these customers approved in the past two years were first denied credit by a major bank.

“The challenge of obtaining credit continues to be very acute for so many,” says Wang. “Using AWS solutions, we are able to keep evolving and growing, knowing our infrastructure is equipped to scale with us.”

About Petal

Petal helps people access and build credit using data already in their banking history. With this data, the company takes into account responsible spending and saving behavior not accounted for in a traditional credit score to make credit as affordable and accessible as possible.

AWS Services Used

Amazon RDS for PostgreSQL

PostgreSQL has become the preferred open source relational database for many enterprise developers and startups, powering leading business and mobile applications. Amazon RDS makes it easier to set up, operate, and scale PostgreSQL deployments on the cloud.

[Learn more »](#)

Amazon SageMaker

Amazon SageMaker is built on Amazon’s two decades of experience developing real-world ML applications, including product recommendations, personalization, intelligent shopping, robotics, and voice-assisted devices.

[Learn more »](#)



Amazon EKS

Amazon Elastic Kubernetes Service (Amazon EKS) is a managed Kubernetes service to run Kubernetes in the AWS cloud and on-premises data centers.

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Amazon Redshift

Amazon Redshift uses SQL to analyze structured and semi-structured data across data warehouses, operational databases, and data lakes, using AWS-designed hardware and machine learning to deliver the best price performance at any scale.

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